

WASP-50b

R-0845

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_181221 · created 2026-07-28T19:55:27+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458473.6554 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.013
Transit depth (Rp/Rs)^2	3.05 +/- 0.44 [%]
Orbital Inclination (inc)	83.94 +/- 0.6
Airmass coefficient 1 (a1)	1.204 +/- 0.018
Airmass coefficient 2 (a2)	-0.125 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	8.2
Transit Duration (day)	0.0717 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.013 vs 0.1404 (deviation 2.13σ)
Duration fit vs published	0.0717 d vs 0.0754 d (deviation 0.49σ)
Mid-time O–C	-1.1 min
Depth SNR	10.8
Residual scatter	1.52 %

Light curve

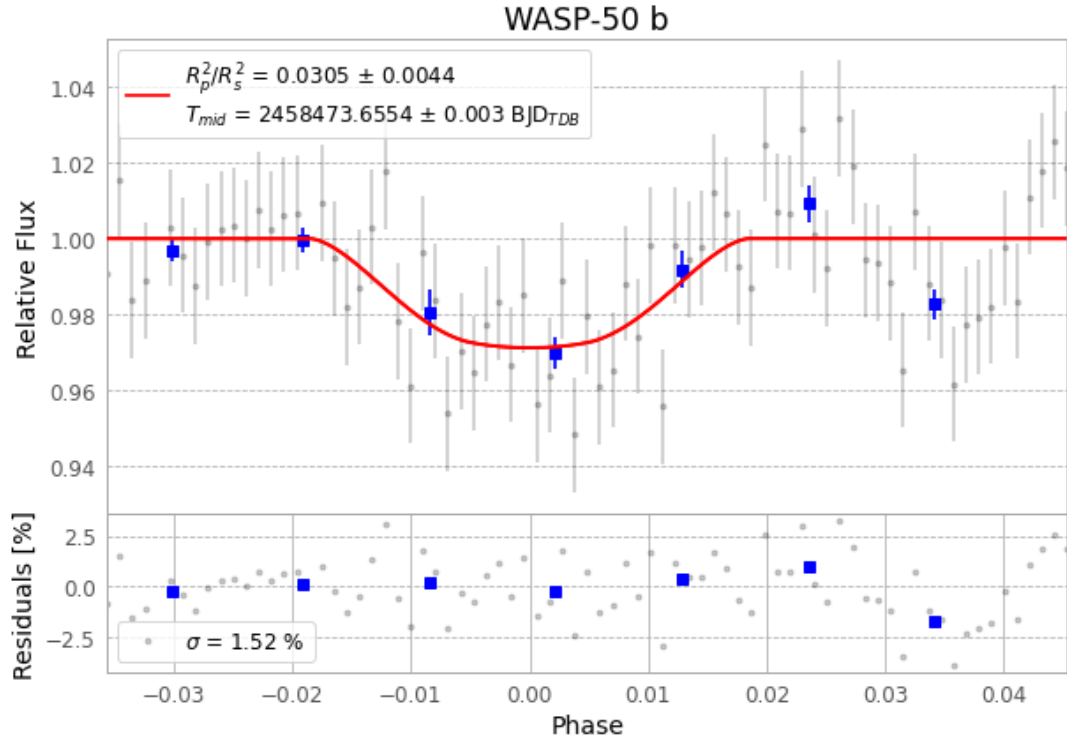


Figure 1 — FinalLightCurve_WASP-50 b_2018-12-20.png (SHA-256 7557fa1b689acad4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad491795ec7f88e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-50181221015712.FITS → WASP-50181221054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-181221.log (0c2a8411cdbc...), FinalLightCurve_WASP-50 b_2018-12-20.png (7557fa1b689a...), FinalLightCurve_WASP-50 b_2018-12-20.csv (3cfada4a1dcc...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 19:55Z.